

AI CODE CONVERTER FOR QUANTUM-CLASSICAL CODE ROUTER

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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I am also grateful to the open-source and quantum computing communities, particularly the teams behind Qiskit, HuggingFace Transformers, and CodeT5, whose tools, pre-trained models, and documentation formed the backbone of my implementation. The availability of the Qiskit Aer simulation framework allowed me to validate generated quantum circuits and performance predictions without requiring access to physical quantum hardware, which was central to the feasibility and scope of this work.

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ABSTRACT

The AI Code Converter is a focused component of the Quantum-Classical Code Router (QCCR) system, designed to transform classical Python algorithms into Qiskit quantum implementations for seven specific algorithm categories. This component addresses the challenge of bridging classical and quantum programming paradigms through machine learning techniques, targeting algorithms that demonstrate clear and well-established quantum advantages.

The system employs a fine-tuned CodeT5 model trained on curated classical-to-quantum algorithm pairs to perform automated code conversion. The seven supported transformations are: Linear Search to Grover's Algorithm, Hidden Bitstring to Bernstein-Vazirani, Parity Check to Deutsch's Algorithm, Swap to Quantum Swap, State Transfer to Quantum Teleportation, Period Finding to Quantum Fourier Transform, and Factoring to Shor's Algorithm. A dedicated Abstract Syntax Tree (AST) analysis module was developed independently to assess the quantum compatibility score of input code, identifying structural patterns that indicate suitability for quantum transformation without relying on graph neural networks. A Performance Prediction module utilizes Qiskit simulators to estimate potential speedup and resource requirements.

The system was validated against benchmark implementations of all seven algorithm categories, achieving strong conversion accuracy and pattern recognition performance. This research contributes to the emerging field of hybrid quantum-classical computing by providing an automated translation layer that lowers the barrier for classical developers to leverage quantum computing advantages.

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LIST OF ABBREVIATIONS

Abbreviation	Description
QCCR	Quantum-Classical Code Router
ML	Machine Learning
API	Application Programming Interface
ASTs	Abstract Syntax Trees
REST	Representational State Transfer

1 INTRODUCTION

The advancement of quantum computing in the Noisy Intermediate-Scale Quantum (NISQ) era has created unprecedented opportunities for solving computationally intensive problems that are intractable for classical systems [1]. However, the practical adoption of quantum computing is significantly hindered by the complexity of quantum programming and the expertise required to transform classical algorithms into quantum equivalents [2].

Current quantum programming requires a deep understanding of quantum mechanics principles, gate-level circuit design, and specialized quantum programming languages such as Qiskit, Cirq, or Q# [3]. This creates a substantial barrier for classical developers who wish to leverage quantum computing advantages without extensive quantum computing expertise.

The field of automated code transformation has gained significant attention as researchers recognise that intelligent code conversion systems can democratise access to quantum computing resources [4]. Major quantum cloud providers including IBM Quantum, Amazon Braket, and Google Quantum AI have made quantum hardware accessible, yet the lack of automated conversion tools limits their practical utility for classical developers [5].

Existing research in quantum programming automation primarily focuses on circuit optimisation and gate synthesis, with limited attention to high-level algorithmic transformation [6]. While several frameworks exist for quantum circuit compilation and optimisation, none provide comprehensive classical-to-quantum conversion capabilities that consider both algorithmic suitability and hardware constraints [7].

Recent advances in Large Language Models and code generation have demonstrated remarkable capabilities in automated programming tasks, providing a strong foundation for intelligent quantum code conversion systems [8]. The integration of pattern recognition through Abstract Syntax Tree analysis and performance prediction enables automated identification of quantum-amenable algorithms and provides developers with actionable quantum advantage estimates.

This research implements a practical AI-powered code converter that fine-tunes CodeT5 on seven well-defined classical-to-quantum algorithm pairs, builds an independent AST-based compatibility scoring system, and integrates a performance prediction module delivering a unified pipeline for quantum code generation that is validated against known quantum implementations.

1.1 Background And Literature Survey

The emergence of quantum computing has introduced a new computational paradigm that promises exponential speedups for specific classes of problems [1]. However, the current quantum computing landscape is dominated by Noisy Intermediate-Scale Quantum (NISQ) devices, which present unique challenges including limited qubit counts, short coherence times, and high error rates [2]. This has created a critical need for hybrid quantum-classical systems that can intelligently distribute computational workloads between classical and quantum processors.

Current approaches to quantum programming require specialized knowledge of quantum mechanics, quantum algorithms, and quantum programming languages such as Qiskit [3], Cirq [4], or Q# [5]. This steep learning curve has created a significant barrier to adoption, limiting the potential impact of quantum computing technologies. Recent research has explored automated quantum circuit synthesis [6] and quantum algorithm compilation [7], but these approaches typically focus on low-level optimisation rather than high-level code conversion.

Machine learning approaches to code generation and transformation have shown remarkable progress, particularly with the advent of large language models trained on code repositories [8]. Models like CodeT5 [9], CodeBERT [10], and GPT-based architectures [11] have demonstrated capabilities in code understanding, generation, and translation between programming languages. CodeT5 is particularly well-suited for this task due to its identifier-aware pretraining on large code corpora, enabling it to understand the semantic meaning of variable names and function signatures, which is critical when mapping classical algorithmic constructs to their quantum counterparts.

Abstract Syntax Tree analysis has long been a foundational technique in program analysis, used extensively for tasks such as code similarity detection, bug localization, and refactoring [12]. In the context of this research, AST analysis provides a structured, language-agnostic representation of classical code that enables systematic identification of algorithmic patterns amenable to quantum transformation. Rather than relying on learned graph representations, the AST-based scoring system in this work applies deterministic structural rules to assess quantum compatibility, offering transparency and interpretability in the pattern recognition process.

The seven algorithm transformations targeted in this research; Linear Search to Grover's, Hidden Bitstring to Bernstein-Vazirani, Parity Check to Deutsch's Algorithm, Swap to Quantum Swap, State Transfer to Quantum Teleportation, Period Finding to Quantum Fourier Transform, and Factoring to Shor's Algorithm represent well-established quantum advantage cases. Each of these has a mathematically proven quantum speedup over its classical counterpart, making them ideal targets for an initial automated conversion system [1], [7].

1.2 Research Gap

Despite significant advances in both quantum computing and machine learning, several critical gaps exist in the current research landscape:

I. Automated Code Conversion

Comprehensive automated conversion from classical Python code to quantum equivalents for well-known algorithm categories, while preserving functional correctness and targeting quantum advantage.

II. Pattern Recognition for Quantum Suitability

Current approaches lack dedicated pattern recognition systems capable of scoring classical algorithmic structures for quantum compatibility at the syntactic and structural level using AST analysis.

III. Performance Prediction with Noise Awareness

Existing quantum simulators focus on ideal quantum systems, lacking practical performance prediction that accounts for NISQ device limitations and noise characteristics.

IV. Multi-task Integration

No system integrates code analysis, AST-based compatibility scoring, LLM-based conversion, and performance prediction into a unified pipeline suitable for developer use.

These gaps represent a significant barrier to the practical adoption of quantum computing, particularly for organizations with existing classical codebases seeking to explore quantum acceleration opportunities.

Research Gap	ChatGpt	Deepseek	IBM	My System
Pattern Recognition for Quantum Suitability	✓	✓	✗	✓
Performance Prediction with Noise Awareness	✗	✗	✓	✓
Multi-modal Integration	✗	✗	✗	✓
Feedback-driven Optimisation	✓	✓	✓	✓
Automated Code Conversion	✓	✓	✗	✓

Table 1: Comparison of the existing research methods and proposed methods

1.3 Research Problem

The primary research problem addressed by the AI Code Converter is:

“How can we develop a system that automatically converts classical computational code to quantum-equivalent implementations for a defined set of algorithm categories, while accurately assessing quantum compatibility and maintaining functional correctness”

This problem encompasses several sub-problems:

1. How to accurately identify quantum-amenable patterns in classical Python code using AST-based structural analysis?
2. How to generate syntactically correct and functionally equivalent quantum code in Qiskit using a fine-tuned CodeT5 model?
3. How to predict quantum performance advantages?
4. How to integrate AST scoring, LLM-based code conversion, and performance prediction into a unified system?
5. How to validate the converted output against known correct quantum implementations across all seven supported algorithm categories?

1.4 Research Objectives

1.4.1 Main objective

The primary objective of the present research is to develop a focused AI-powered code converter that automatically transforms classical Python implementations of seven defined algorithm categories into Qiskit quantum implementations, supported by an AST-based compatibility scoring system.

1.4.2 Specific sub objectives

I. Large Language Model Implementation

The CodeT5-base model was fine-tuned on a curated dataset of classical-to-quantum algorithm pairs, specifically targeting Python-to-Qiskit translation across seven well-defined algorithm categories: Linear Search to Grover's Algorithm, Hidden Bitstring to Bernstein-Vazirani, Parity Check to Deutsch's Algorithm, Swap to Quantum Swap, State Transfer to Quantum Teleportation, Period Finding to Quantum Fourier Transform, and Factoring to Shor's Algorithm. The fine-tuning process focused on domain specificity and model compactness, with correctness validated through structured benchmarks and comparison against known quantum implementations.

II. AST-Based Pattern Recognition System Development

An independent AST-based pattern recognition and compatibility scoring engine was developed to assess whether a given classical Python algorithm is a candidate for quantum transformation. The system parses input code into an Abstract Syntax Tree, extracts structural features such as loop constructs, conditional branches, arithmetic operations, and function call patterns, and maps these against a pattern database of known quantum-amenable algorithmic signatures for the seven supported categories. This module operates independently of the LLM conversion pipeline and produces a compatibility score that informs the user of the suitability of their code for quantum transformation. No graph neural network component was used; the scoring logic is fully AST-driven.

III. Performance Prediction Module

Using Qiskit Aer simulators and basic noise models, the expected performance of quantum-translated code was estimated under NISQ conditions. The module provides qubit usage estimates, gate depth analysis, and rough runtime comparisons against classical counterparts for each of the seven supported algorithm types, allowing users to assess the practical quantum advantage of converting their code.

IV. System Integration and Validation

The AST scoring engine, CodeT5 conversion model, and performance prediction module were integrated into a unified backend with a REST API and a simple web interface. The full system was validated using benchmark implementations of all seven supported algorithm categories, comparing generated Qiskit outputs against known correct manual implementations to confirm functional equivalence and conversion quality.

2 METHODOLOGY

2.1 METHODOLOGY

2.1.1 Overall System Architecture

The AI Code Converter serves as an intelligent translation module within the broader Quantum-Classical Code Router (QCCR) system. It receives classical Python code from the user and processes it through three sequential components: an AST-based compatibility scoring engine, a fine-tuned CodeT5 conversion model, and a Qiskit-based performance prediction module. The converted quantum code, along with compatibility scores and performance estimates, is returned to the user and forwarded to the QCCR Decision Engine for hardware routing decisions.

The overall flow is as follows: the user submits classical Python code through a web interface or REST API. The code is first passed to the AST Analysis Module, which parses the code structure and produces a quantum compatibility score along with an identified algorithm category. If the code is deemed compatible, the user can use the CodeT5 LLM Layer, to generate the corresponding Qiskit quantum implementation. The generated code is then evaluated by the Performance Prediction Module using Qiskit Aer simulators to estimate qubit requirements, gate depth, and potential speedup. The final output the quantum code, compatibility score, and performance report is returned to the user.

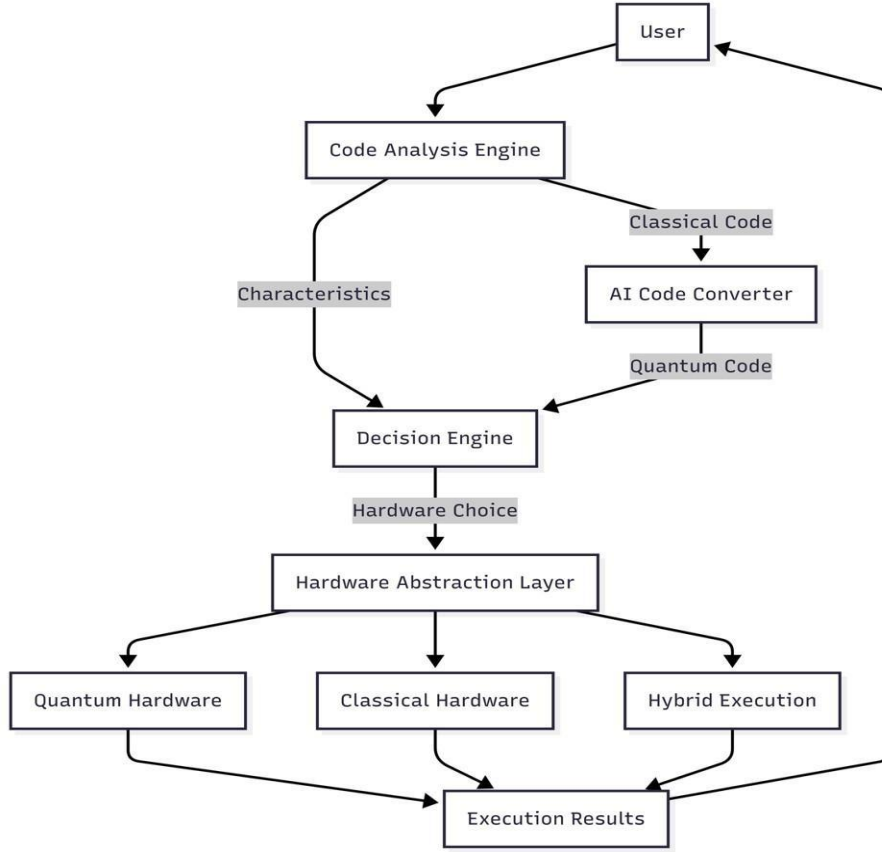


Figure 1: Complete System Architecture

2.1.2 AI-Code Converter System Architecture

The high-level system architecture consists of the following components:

User Interface / Code Input Module: A simple web-based frontend that accepts classical Python code input and displays the converted quantum code, compatibility score, and performance prediction results.

Conversion Request Handler: A REST API layer that receives code submission requests, routes them through the processing pipeline, and returns structured results to the interface.

AST Pattern Recognition System: The input code is parsed into an Abstract Syntax Tree. Structural features are extracted and compared against a pattern database of seven algorithm signatures. A compatibility score is computed based on the presence and arrangement of relevant syntactic constructs. This module is entirely separate from the LLM layer and operates deterministically.

Large Language Model Layer (CodeT5): The fine-tuned CodeT5-base model receives the classical Python code and generates a Qiskit quantum implementation. The model was fine-tuned specifically on the seven supported algorithm transformation pairs and produces syntactically valid Qiskit code as output.

Performance Prediction Module: The generated quantum code is executed on Qiskit Aer simulators with basic noise models applied. Metrics including qubit count, circuit depth, gate composition, and estimated runtime are extracted and compared against classical performance baselines for the identified algorithm category.

Integration and Communication Layer: A REST API connects all internal components and provides the interface through which the broader QCCR system (Code Analysis Engine, Decision Engine, Hardware Abstraction Layer) communicates with the AI Code Converter.

2.1.3 Requirements Gathering And Analysis

The requirements gathering and analysis phase was conducted through a combination of literature review, technical scoping, and systematic validation using synthetically generated test data. Given the novelty of classical-to-quantum code conversion as a domain, traditional user consultation was supplemented by a deep review of established quantum algorithm literature to define the functional boundaries of the system.

The first step involved identifying which classical algorithm categories have well-proven, deterministic quantum counterparts with established speedup advantages. This narrowed the scope to seven transformation pairs: Linear Search to Grover's Algorithm, Hidden Bitstring to Bernstein-Vazirani, Parity Check to Deutsch's Algorithm, Swap to Quantum Swap, State Transfer to Quantum Teleportation, Period Finding to Quantum Fourier Transform, and Factoring to Shor's Algorithm. Each of these pairs represents a case where the quantum advantage is mathematically established and the mapping between classical and quantum constructs is well-defined, making them suitable targets for automated conversion and validation.

Because real-world datasets of paired classical-to-quantum code are extremely scarce in the literature, a synthetic data generation strategy was adopted for both training and testing. For the fine-tuning of CodeT5 and for validating the AST scoring engine, synthetic classical Python code samples were programmatically generated for each of the seven algorithm categories. These samples were designed to capture a range of stylistic variations - including differences in variable naming conventions, loop structures, function decomposition styles, and input parameter formats while preserving the core algorithmic logic of each category. Each synthetic classical sample was paired with a manually verified correct Qiskit implementation of the corresponding quantum algorithm, producing a labelled dataset of classical-quantum code pairs.

The synthetic samples were reviewed for correctness and diversity before being used for fine-tuning and evaluation. For example, for the Linear Search to Grover's Algorithm transformation, classical samples included variations using for loops with explicit return statements, while loops with index tracking, and list comprehension-style search patterns. The corresponding Qiskit implementations consistently used Grover's oracle construction, amplitude amplification, and measurement circuits as per

established quantum algorithm design. A similar approach was applied across all seven categories, resulting in a curated dataset that covers the structural diversity a developer might realistically present to the system.

Testing was also conducted using synthetic inputs that were deliberately designed to be edge cases inputs with unusual but valid Python constructs, inputs that superficially resemble a supported category but belong to a different one, and inputs that are entirely outside the seven supported categories. These edge case tests were used to validate the robustness of the AST compatibility scoring system, confirming that it correctly assigns low compatibility scores to unsupported patterns and does not produce false positives that would lead to incorrect conversion attempts.

The requirements gathered through this process informed the definition of functional and non-functional requirements, scoped the training and evaluation datasets, and established the benchmark criteria against which the final system was validated.

2.1.3.1 Feasibility Study

The feasibility study is a critical and foundational phase in the development lifecycle of any software system. It plays a vital role in identifying whether the proposed solution is viable in terms of available resources, technology, and real-world applicability. For this research focused on the development of an integrated diabetic management system, the feasibility study is centered on thoroughly evaluating two primary dimensions technical feasibility and operational feasibility to determine whether the proposed system can be successfully developed, implemented, and sustained in actual healthcare environments.

2.1.3.1.1 Technical Feasibility

The technical feasibility of the AI Code Converter was assessed by examining whether the proposed combination of a fine-tuned language model, an AST-based analysis module, and a quantum simulation environment could be successfully implemented using available open-source tools and standard computing infrastructure.

The core machine learning component relies on CodeT5-base, a transformer-based encoder-decoder model available through the HuggingFace Transformers library. CodeT5 has been widely validated for code-to-code translation tasks and is well-supported by an active ecosystem of training utilities, tokenizers, and evaluation frameworks. Fine-tuning CodeT5 on the curated classical-to-quantum dataset is technically straightforward using the HuggingFace datasets and evaluate libraries alongside PyTorch and the accelerate library for training efficiency. The model's relatively compact base size makes it suitable for fine-tuning on standard GPU hardware without requiring large-scale distributed compute infrastructure.

The AST-based pattern recognition component is implemented entirely using Python's built-in `ast` module, which provides a stable and well-documented interface for parsing Python source code into a structured tree representation. No external graph processing libraries or neural network components are required for this module, making it lightweight, deterministic, and straightforward to maintain. The `astor` library was additionally used for AST-to-source-code reconstruction where needed during analysis.

The quantum simulation component relies on Qiskit and Qiskit Aer, which are the most widely used open-source quantum computing frameworks available. Qiskit provides comprehensive tools for constructing, transpiling, and simulating quantum circuits, while Qiskit Aer provides high-performance simulation backends. Both libraries are actively maintained by IBM and have extensive documentation and community support. The `pylatexenc` library was included to support circuit diagram rendering where required.

The backend API is implemented using FastAPI and Uvicorn, which provide a modern, high-performance, asynchronous web framework well-suited for exposing machine learning model inference and simulation endpoints. FastAPI's automatic request validation through Pydantic and its OpenAPI documentation generation make it particularly suitable for integration with the broader QCCR system components. The `httpx` and `requests` libraries support inter-service communication, and `pytest` provides a structured framework for automated testing of all API endpoints and internal components.

Data processing, evaluation, and performance visualisation are supported by NumPy, Pandas, Matplotlib, SciPy, and Seaborn, which together provide a complete environment for numerical analysis, result aggregation, and chart generation. Cloud deployment and logging are supported through Google Cloud Storage and Google Cloud Logging integrations.

Overall, every component of the system is built on mature, well-supported open-source technology. No proprietary systems, specialised hardware beyond a standard GPU, or unproven frameworks are required. The system is technically feasible and was successfully implemented within the available development environment.

2.1.3.1.2 Operational Feasibility

The operational feasibility of the AI Code Converter is assessed in terms of how realistically the system can function as a developer tool within the broader QCCR pipeline and as a standalone utility for classical developers exploring quantum computing.

The system is designed with a clear separation of concerns between its three core components the AST scoring engine, the CodeT5 conversion model, and the performance prediction module each exposed through well-defined REST API endpoints. This modular architecture ensures that the system integrates naturally into

the QCCR system's existing component structure and can be independently updated or extended without disrupting the broader pipeline.

From a usability perspective, the web interface accepts standard Python code as plain text input, requiring no special formatting or quantum computing knowledge from the user. The output is structured to provide the user with the identified algorithm category, the quantum compatibility score, the generated Qiskit code, and a performance summary all in a single response. This design ensures that the system is accessible to classical developers who may have no prior experience with quantum programming.

The scope of the system is deliberately bounded to seven well-defined algorithm categories, which makes the operational behaviour of the system predictable and its outputs verifiable. Users and system operators can clearly understand what the system supports and what falls outside its current capabilities, reducing the risk of misuse or misinterpretation of results.

The system was validated operationally by testing all seven algorithm categories end-to-end through the full pipeline, from code submission through the web interface to the return of quantum code and performance metrics, confirming that the integrated system behaves as expected under normal operating conditions.

2.1.3.2 Tools, Technologies & Algorithms

2.1.3.2.1 Backend Framework

The system backend is built using FastAPI (0.104.1) with Uvicorn (0.24.0) as the ASGI server. FastAPI was selected for its high performance, native asynchronous request handling, automatic OpenAPI documentation generation, and tight integration with Pydantic for request and response validation. Pydantic (2.5.0) is used throughout the API layer to define and enforce strict data schemas for code submission requests and conversion responses. Python-multipart supports multipart form data handling for code file uploads, and python-dotenv manages environment variable configuration across development and deployment environments.

2.1.3.2.2 Quantum Computing Framework

Qiskit serves as the primary quantum computing framework, used for constructing all seven quantum circuit implementations generated by the conversion pipeline. Qiskit Aer provides the simulation backend, offering both ideal statevector simulation and noise-model-based simulation for NISQ device emulation. The Aer simulator was used in the performance prediction module to estimate qubit counts, gate depths, and execution characteristics for each of the seven supported quantum algorithm types.

Pylatexenc is included to support LaTeX-based quantum circuit diagram rendering where circuit visualisation is required.

2.1.3.2.3 Machine Learning and Model Fine-Tuning

HuggingFace Transformers provides the CodeT5-base model architecture and tokenizer used for the classical-to-quantum code conversion component. The model was fine-tuned using PyTorch as the underlying deep learning framework. The datasets library was used to structure and load the synthetic classical-to-quantum training pairs, and the evaluate library was used to compute BLEU and other code generation metrics during fine-tuning evaluation. Accelerate ($\geq 0.26.0$) was used to streamline the training loop and support efficient GPU utilisation during fine-tuning. Sentencepiece and protobuf provide tokenisation and serialisation support required by the Transformers library.

2.1.3.2.4 AST Analysis

Python's built-in ast module forms the core of the pattern recognition and compatibility scoring engine. Input Python code is parsed into an Abstract Syntax Tree, and a custom traversal pipeline extracts structural features relevant to each of the seven supported algorithm categories. The astor library was used for AST-to-source reconstruction tasks during analysis and debugging of the pattern matching logic.

2.1.3.2.5 Data Processing and Evaluation

NumPy and Pandas are used for numerical computation, dataset management, and result aggregation throughout the training, evaluation, and performance prediction pipeline. Scikit-learn provides evaluation utilities including classification metrics used to assess the AST pattern recognition system's accuracy across the seven algorithm categories. SciPy supports statistical analysis of prediction results, and Matplotlib and Seaborn are used to generate performance and evaluation visualisations.

2.1.3.2.6 Testing and Communication

Pytest provides the automated testing framework used for unit and integration testing of all system components. Requests and HTTPX support HTTP client functionality for inter-service communication and for programmatic API testing alongside Postman-

based manual endpoint validation. Pillow provides image processing support for any circuit diagram output handling.

2.1.3.2.7 Cloud Infrastructure

Google Cloud Storage is used for storing trained model artefacts and dataset files, and Google Cloud Logging provides centralised logging for deployment monitoring and debugging.

2.1.3.2.8 Algorithms

The seven quantum algorithms implemented as conversion targets are as follows. Grover's Algorithm provides quadratic speedup for unstructured search, replacing classical linear search with an amplitude amplification circuit that identifies the target state in $O(\sqrt{N})$ oracle calls. Bernstein-Vazirani Algorithm solves the hidden bitstring problem in a single query using quantum superposition and interference, compared to N classical queries. Deutsch's Algorithm determines the parity of a binary function in a single evaluation using quantum parallelism, replacing a two-query classical approach. Quantum Swap implements the exchange of two quantum states using three CNOT gates, demonstrating fundamental quantum gate operations. Quantum Teleportation transfers an arbitrary qubit state between locations using an entangled Bell pair and classical communication, replacing classical state copying with a quantum protocol. Quantum Fourier Transform provides the quantum analogue of the Discrete Fourier Transform with exponential speedup, and forms the basis for Period Finding conversion. Shor's Algorithm factors integers in polynomial time using quantum period finding combined with classical number theory, replacing exponential-time classical factoring algorithms.



Figure 2: Used Tools and Technologies

2.1.4 Project Requirements

2.1.4.1 Functional requirements

Functional requirements define the specific behaviours, capabilities, and operations that the AI Code Converter system must perform. These requirements were derived from the research objectives, the synthetic dataset validation process, and the operational needs of the broader QCCR system integration.

FR1: Code Analysis and Pattern Recognition

The first functional requirement category covers the system's ability to analyse submitted classical Python code and determine its suitability for quantum transformation through AST-based structural analysis.

FR1.1: Abstract Syntax Tree Parsing

The system must accept raw classical Python source code as input and successfully parse it into an Abstract Syntax Tree representation using Python's built-in ast module. This parsing step is the foundation of the entire analysis pipeline. The system must handle standard Python syntax correctly and return a structured error response if the submitted code cannot be parsed due to syntax errors, rather than passing malformed input downstream to the conversion model. The AST representation must capture all relevant structural elements of the submitted code including function definitions, loop constructs, conditional branches, arithmetic and bitwise operations, variable assignments, and function call patterns.

FR1.2: Algorithm Category Identification

Using the extracted AST features, the system must identify which of the seven supported algorithm categories the input code most closely matches. The seven supported categories are Linear Search, Hidden Bitstring, Parity Check, Swap, State Transfer, Period Finding, and Factoring. The identification process works by comparing the structural feature profile of the input code against a pattern database

containing the characteristic AST signatures of each category. The system must return the identified category name as part of its response so that the user and downstream QCCR components are aware of which transformation was applied. If the input does not match any supported category above the defined compatibility threshold, the system must report this clearly rather than proceeding with an incorrect conversion.

FR1.3: Quantum Compatibility Scoring

Beyond binary category identification, the system must produce a continuous quantum compatibility score between zero and one that reflects how closely the structural profile of the input code matches the expected pattern for the identified category. This score is computed as a weighted function of the presence of required structural features, the presence of optional supporting features, and the absence of disqualifying features for each category. A score close to one indicates a strong structural match with high confidence that the conversion will produce a correct and meaningful quantum implementation. A score significantly below one indicates that while the code may belong to a supported category, its structure deviates from the expected pattern in ways that may affect conversion quality. This score is reported to the user and logged for system monitoring purposes.

FR2: Code Conversion Capabilities

The second functional requirement category covers the system's ability to generate correct and functional Qiskit quantum code from classical Python input using the fine-tuned CodeT5 model.

FR2.1: Syntactically Correct Qiskit Code Generation

The system must generate Qiskit quantum code that is syntactically valid and executable within the Qiskit framework for all seven supported algorithm categories. This means the generated code must correctly use Qiskit's QuantumCircuit API, apply gates using valid method calls such as `circuit.h()`, `circuit.cx()`, `circuit.measure()`, and correctly declare quantum and classical registers with appropriate sizes. The generated code must be directly executable in a standard Python environment with Qiskit installed, without requiring manual correction of syntax errors or API usage mistakes by the user.

FR2.2: Functional Equivalence Between Classical and Quantum Implementations

The generated quantum code must be functionally equivalent to the submitted classical code in the sense that it solves the same computational problem. For each of the seven supported categories, functional equivalence is defined specifically: for Linear Search to Grover's, the quantum circuit must identify the same target element that the classical search would find. For Hidden Bitstring to Bernstein-Vazirani, the circuit must recover the same hidden bitstring that the classical dot product computation would determine. For Parity Check to Deutsch's, the circuit must correctly classify the function as constant or balanced in the same way the classical evaluation would. For Swap to Quantum Swap, the circuit must exchange the two input states in the same way the classical swap operation exchanges variable values. For State Transfer to Teleportation, the circuit must transfer the input qubit state to the target register. For Period Finding to QFT, the circuit must identify the same period that classical modular exponentiation would find. For Factoring to Shor's, the circuit must produce period measurements consistent with the correct factors of the input integer. Functional equivalence is validated by executing generated circuits on the Qiskit Aer simulator and verifying output distributions against expected results.

FR2.3: Support for All Seven Transformation Categories

The system must support all seven of the following classical-to-quantum code transformation types without exception, as these represent the complete scope of the current system:

- Linear Search → Grover's Algorithm
- Hidden Bitstring → Bernstein-Vazirani Algorithm
- Parity Check → Deutsch's Algorithm
- Swap → Quantum Swap
- State Transfer → Quantum Teleportation
- Period Finding → Quantum Fourier Transform
- Factoring → Shor's Algorithm

Each transformation must be independently functional, meaning a failure or degraded performance in one category must not affect the system's ability to correctly process inputs from the other six categories. The system must also clearly communicate to the user which transformation was applied in the conversion response.

FR3: Performance Prediction and Analysis

The third functional requirement category covers the system's ability to simulate generated quantum circuits and produce meaningful performance estimates that help users understand the practical implications of converting their code to quantum.

FR3.1: Quantum vs Classical Performance Comparison

For each of the seven supported algorithm categories, the system must produce an estimate of the quantum performance advantage relative to the classical counterpart. For categories with well-established asymptotic speedups specifically Grover's Algorithm (quadratic speedup over linear search), Bernstein-Vazirani (exponential reduction in query complexity), Deutsch's Algorithm (single query versus two classical queries), and Shor's Algorithm (polynomial versus exponential factoring time) the advantage estimate is computed analytically based on the input size and reported alongside the simulation results. For the remaining categories, the comparison is presented in terms of circuit resource usage relative to the classical operation count.

FR3.2: Circuit Resource Estimation

The system must extract and report the following resource metrics for every generated quantum circuit: the total number of qubits required, the total number of quantum gates in the circuit before and after transpilation to a hardware basis gate set, the circuit depth after transpilation, and the estimated execution time on a representative NISQ simulator. These metrics are extracted from the Qiskit transpiler output and Aer simulation run data. This information allows users to assess whether their algorithm is within the operational range of current or near-term quantum hardware, and allows the QCCR Decision Engine to make informed hardware routing choices.

FR3.4: Performance Report Generation

The system must compile all performance metrics qubit count, gate count, circuit depth, ideal fidelity, noise-model fidelity, and quantum advantage estimate into a structured performance report that is returned as part of the API response. This report must be human-readable and must clearly communicate the key takeaway for each metric, such as whether the quantum advantage is practically realisable on current

hardware given the circuit depth and noise characteristics, or whether the algorithm would benefit more from execution on a future fault-tolerant quantum device.

FR4: Validation and Verification

The fourth functional requirement category covers the system's internal mechanisms for confirming that generated outputs are correct and complete before returning them to the user.

FR4.1: Quantum Simulator Validation

Every quantum circuit generated by the CodeT5 conversion model must be validated by executing it on the Qiskit Aer statevector simulator before being returned to the user. The simulation must complete without runtime errors, and the output measurement distribution must be checked against the expected result pattern for the identified algorithm category. If the simulation fails or produces an output that does not match the expected pattern for example, if a Grover's circuit does not show a dominant amplitude at any state the system must flag this in the response rather than silently returning a potentially incorrect circuit.

FR4.2: Functional Equivalence Checking

For each conversion, the system must perform an automated functional equivalence check by running both the classical input logic and the generated quantum circuit on equivalent test inputs and comparing their outputs. For deterministic algorithms such as Deutsch's, Bernstein-Vazirani, and Swap, exact output matching is checked. For probabilistic algorithms such as Grover's and Shor's, the check verifies that the correct answer appears with high probability in the measurement output distribution. This equivalence check result is included in the conversion report returned to the user.

FR4.3: Detailed Conversion Report

The system must return a comprehensive conversion report for every processed request. This report must include the identified algorithm category, the quantum

compatibility score from the AST analysis, the generated Qiskit source code, the circuit resource metrics, the ideal and noise-model simulation results, the functional equivalence check result, and a plain-language summary of the quantum advantage assessment for the identified algorithm type. This report provides the user with all the information needed to understand and use the generated quantum implementation confidently.

FR5: System Integration and Communication

The fifth functional requirement category covers the system's interfaces with external components, including the broader QCCR system and end users.

FR5.1: RESTful API for QCCR Integration

The system must expose a RESTful API that allows the other components of the Quantum-Classical Code Router system to interact with the AI Code Converter programmatically. This includes a compatibility scoring endpoint that accepts Python code and returns the AST analysis results for use by the QCCR Code Analysis Engine, a conversion endpoint that accepts Python code and returns the generated Qiskit implementation for use by the Decision Engine, and a full pipeline endpoint that runs all stages and returns a complete report. All endpoints must accept and return JSON-formatted data, implement appropriate HTTP status codes for success and error conditions, and include basic authentication to prevent unauthorised access.

FR5.2: Web Interface for Developer Use

The system must provide a simple web-based interface that allows developers to submit classical Python code and receive the conversion output, compatibility score, and performance report without requiring knowledge of the API or quantum computing tooling. The interface must include a code input area, a submission control, and clearly organised output panels for the AST analysis result, the generated Qiskit code, and the performance summary. The interface must be functional in standard modern web browsers without requiring any client-side installation.

2.1.4.2 Non-functional requirements

Non-functional requirements define the quality attributes, operational constraints, and system characteristics that govern how the AI Code Converter performs its functions. These requirements address performance, reliability, usability, security, and maintainability across the full scope of the system.

NFR1: Performance Requirements

NFR1.1: Conversion Accuracy

The system must achieve high conversion accuracy across all seven supported algorithm transformation categories. Accuracy is defined in terms of the proportion of test inputs for which the generated Qiskit code passes both the syntactic validity check and the functional equivalence validation against the reference quantum implementation. The fine-tuned CodeT5 model must produce correct and complete circuit structures for the large majority of test inputs across all seven categories, with the strongest performance expected for structurally regular transformations such as Grover's, Bernstein-Vazirani, and Deutsch's, and acceptable performance for the more complex transformations such as Shor's Algorithm and QFT.

NFR1.2: AST Pattern Recognition Accuracy

The AST-based pattern recognition engine must correctly identify the algorithm category for the large majority of supported inputs across all seven categories. The engine must maintain a low false positive rate for unsupported inputs, ensuring that code outside the supported categories is not incorrectly assigned to a supported category and passed to the conversion model. Compatibility scores assigned by the engine must be consistent and meaningful, with high scores reliably predicting successful conversion and low scores reliably indicating poor conversion candidates.

NFR1.3: System Response Time

The system must complete the full pipeline AST analysis, CodeT5 inference, Qiskit simulation, and report generation within a response time that is acceptable for interactive developer use. The AST analysis stage must complete in well under one second for any reasonable input size. The CodeT5 inference stage is expected to take longer due to model inference overhead but must complete within a timeframe that does not make the system impractical to use interactively. The Qiskit Aer simulation stage execution time will vary with circuit depth and qubit count but must complete

within an acceptable window for the circuit sizes generated by the seven supported algorithm categories.

NFR1.4: Performance Prediction Consistency

The performance prediction module must produce consistent results across repeated runs on the same input. Qubit counts and gate depths must be deterministic for a given generated circuit. Noise-model simulation results may vary slightly due to the stochastic nature of quantum measurement simulation but must produce output distributions that are statistically consistent with expected results across repeated runs.

NFR2: Scalability and Reliability

NFR2.1: Concurrent Request Handling

The FastAPI backend must handle multiple concurrent conversion requests without request queuing failures or response corruption. The asynchronous architecture of FastAPI and Uvicorn provides the foundation for concurrent request handling, and the system must be configured to support a reasonable number of simultaneous requests without degradation in response quality. Individual requests must be processed independently, meaning a long-running Qiskit simulation for one request must not block or delay the AST analysis stage of another concurrent request.

NFR2.2: Stability and Uptime

The system must operate stably over extended periods without requiring manual restarts due to memory leaks, model state corruption, or unhandled exceptions. All API endpoints must implement comprehensive exception handling that catches and logs errors at every stage of the pipeline AST parsing, model inference, Qiskit circuit construction, and simulation and returns a structured error response to the client rather than an unhandled server error. The system must maintain high availability during normal operating conditions.

NFR2.3: Error Handling and Logging

The system must implement structured logging at all stages of the processing pipeline using Google Cloud Logging in deployment. Every request must be logged with a unique request identifier, the submitted code length, the identified algorithm category, the compatibility score, the conversion outcome, and the total processing time. Errors

at any stage must be logged with sufficient detail to support debugging, including the stage at which the error occurred, the error type, and relevant context information. This logging infrastructure ensures that system behaviour can be monitored and analysed over time, and that issues can be diagnosed efficiently.

NFR3: Usability and Maintainability

NFR3.1: Developer-Facing Interface Clarity

The web interface must present its inputs and outputs in a clear and well-organised manner that does not require quantum computing expertise to interpret. The compatibility score must be displayed with a plain-language explanation of what it means. The identified algorithm category must be clearly labelled with the specific transformation that was applied. The generated Qiskit code must be displayed with syntax highlighting in a scrollable code panel. The performance summary must present key metrics such as qubit count, circuit depth, and quantum advantage estimate in plain language rather than raw numerical output alone. A user with a background in classical Python programming but no prior quantum computing experience must be able to understand the output of the system and make use of the generated quantum code with the help of the provided report.

NFR3.2: Output Format Support

The system must support returning the conversion report in a structured JSON format via the API for programmatic consumption by the QCCR Decision Engine and other system components. The performance summary must include all key metrics in clearly named JSON fields. The generated Qiskit source code must be returned as a plain string that can be directly written to a Python file and executed. Where circuit diagram rendering is supported, the diagram must be returned as a base64-encoded image or a rendered PNG file.

NFR3.3: Modular Architecture for Maintainability

The AST scoring engine, the CodeT5 fine-tuning and inference pipeline, and the Qiskit simulation and performance prediction module must be implemented as independent, loosely coupled modules with clearly defined interfaces between them. This modularity ensures that any one component can be updated, retrained, or replaced without requiring changes to the other components. For example, the CodeT5 model can be retrained on an expanded dataset to support additional algorithm categories without modifying the AST scoring logic or the performance prediction module. Similarly, the AST pattern database can be extended with new algorithm signatures

without affecting the LLM layer. This separation of concerns is essential for the long-term maintainability and extensibility of the system.

NFR3.4: System Documentation

The system must be accompanied by comprehensive technical documentation covering the API endpoint specifications, the AST pattern database schema, the CodeT5 fine-tuning procedure, the Qiskit simulation configuration, and the deployment process. A user guide must be provided that explains how to submit code to the system, how to interpret the compatibility score and conversion report, and how to execute the generated Qiskit code in a local or cloud quantum computing environment. This documentation ensures that the system can be operated, maintained, and extended by developers who were not involved in its original implementation.

NFR4: Security and Compliance

NFR4.1: API Authentication and Authorisation

All REST API endpoints must be protected by authentication mechanisms that prevent unauthorised access. API key-based authentication must be implemented at the API gateway level, with keys managed through environment variables loaded via python-dotenv. Requests without valid authentication credentials must be rejected with an appropriate HTTP 401 or 403 response before any processing is performed. Within the QCCR system, the AI Code Converter endpoints must only be accessible to authorised QCCR components, enforcing the principle of least privilege across the system architecture.

NFR4.2: Code Confidentiality

Submitted source code must be treated as potentially sensitive intellectual property. The system must not persist submitted code to any storage medium beyond the duration of the processing request. Code submitted through the API must be processed entirely in memory and discarded after the response is returned. Logs must record metadata about requests such as code length, identified category, and processing time but must not record the submitted source code itself. This ensures that users can submit proprietary algorithms without risk of their code being retained or exposed.

NFR4.3: Cloud Security Standards

When deployed on Google Cloud Platform, the system must adhere to standard cloud security practices including the use of service account credentials with minimal required permissions for Google Cloud Storage and Google Cloud Logging access, encrypted data transmission over HTTPS for all API endpoints, and secure management of all credentials and API keys through environment variables rather than hardcoded values in source code. The deployment configuration must be reviewed to ensure compliance with relevant cloud security standards before production use.

2.2 Commercialization Aspects Of The Product

The AI Code Converter demonstrates considerable commercial potential within the rapidly expanding quantum computing services market. As quantum hardware becomes increasingly accessible through cloud platforms such as IBM Quantum, Amazon Braket, and Google Quantum AI, the limiting factor for adoption is no longer hardware availability but developer accessibility. Organisations with established classical Python codebases have limited means to explore quantum acceleration without investing heavily in quantum computing expertise. The AI Code Converter directly addresses this gap by providing an automated translation layer that requires no quantum programming knowledge from the end user, making it commercially viable as a developer productivity tool in the quantum computing transition period.

The most immediate commercialisation pathway is integration into the broader Quantum-Classical Code Router system as a paid component of a hybrid quantum-classical computing platform. In this model, the QCCR system as a whole including the AI Code Converter, Decision Engine, and Hardware Abstraction Layer could be offered as a managed cloud service to enterprise clients who wish to evaluate quantum acceleration opportunities within their existing software infrastructure. The AI Code Converter would function as the entry point of this service, processing classical code submissions and routing converted quantum implementations to appropriate hardware through the QCCR pipeline. Enterprise pricing in this model could be structured around a per-conversion billing model, a subscription tier based on monthly conversion volume, or a platform licensing model for organisations deploying the system within their own infrastructure.

A second commercialisation pathway is the deployment of the AI Code Converter as a standalone developer tool, offered through a subscription-based Software-as-a-Service model. Individual developers, research teams, and academic institutions could subscribe to access the conversion API and web interface without requiring the full QCCR stack. This model is well-suited to the research and education market, where quantum computing courses and academic projects could benefit significantly from automated classical-to-quantum translation without requiring students and researchers to develop deep Qiskit expertise. A freemium tier offering a limited number of monthly conversions could serve as an entry point for individual users, with paid tiers unlocking higher conversion volumes, priority processing, and access to detailed performance prediction reports.

The system's modular architecture also creates a licensing opportunity for quantum hardware providers and quantum cloud platforms seeking to lower the barrier to adoption for their services. IBM Quantum, Amazon Braket, and similar platforms have a direct commercial interest in tools that increase the volume and diversity of quantum workloads submitted to their hardware. The AI Code Converter could be licensed to these platforms for integration into their developer portals, providing a value-added service that encourages classical developers to explore quantum execution without requiring them to write Qiskit code manually. This B2B licensing model would generate revenue through a combination of upfront licensing fees and usage-based royalties tied to the volume of quantum circuit executions generated through the conversion pipeline.

The current scope of seven algorithm categories represents a deliberately conservative initial implementation chosen to ensure conversion quality and functional correctness. From a commercial development perspective, this scope can be progressively expanded to include additional algorithm categories such as quantum machine learning circuits, variational quantum eigensolvers, and quantum chemistry algorithms as the training dataset grows and the fine-tuning pipeline matures. Each expansion of the supported algorithm set represents an increase in the addressable market for the tool, extending it from the current focus on search and optimisation algorithms to the broader quantum algorithm landscape.

Future development could also incorporate direct integration with popular classical development environments and version control platforms. A Visual Studio Code extension or a GitHub Action that automatically analyses committed Python code for quantum conversion opportunities, flags compatible algorithms, and optionally generates Qiskit alternatives as pull request suggestions would represent a high-value developer tool with strong commercial appeal. This integration model positions the AI Code Converter not as a standalone application but as an embedded layer within existing classical development workflows, significantly lowering the friction of adoption.

In summary, the AI Code Converter is commercially viable as a component of an enterprise quantum-classical computing platform, as a standalone SaaS developer tool, and as a licensed integration within quantum cloud provider ecosystems. Its core value proposition removing the quantum programming expertise barrier for classical developers is well aligned with the current state of the quantum computing industry, where hardware capability is advancing faster than developer adoption.

2.3 Testing & Implementation

2.3.1 Implementation process

The implementation process of the QCCR System demonstrates a wellstructured and methodical approach, integrating data engineering with machine learning techniques to support translations. Several critical aspects of the process deserve further discussion in terms of their effectiveness and impact on the overall system performance.

2.3.1.1 Dataset Construction

The dataset used for fine-tuning the CodeT5 model and evaluating the AST scoring engine was constructed synthetically due to the absence of large-scale publicly available paired classical-to-quantum code datasets. For each of the seven supported algorithm transformation categories, a set of classical Python implementations was programmatically generated to capture realistic stylistic variation while preserving the core algorithmic logic of each category.

For the Linear Search to Grover's transformation, classical samples included implementations using indexed for loops, while loops with explicit break conditions, and enumerate-based iteration patterns. For the Hidden Bitstring to Bernstein-Vazirani transformation, classical samples implemented bitwise AND-based dot product computation across a range of bitstring lengths. For the Parity Check to Deutsch's transformation, classical samples included explicit if-else parity checks and XOR-based parity functions. Swap implementations varied between direct triple-assignment swap, temporary variable swap, and tuple unpacking swap. State transfer samples implemented classical data copying and serialisation patterns. Period finding samples implemented classical modular exponentiation loops, and factoring samples included trial division, Pollard's rho, and basic brute-force factoring implementations.

Each classical sample was paired with a manually verified and tested Qiskit implementation of the corresponding quantum algorithm. All Qiskit implementations were verified by executing them on the Qiskit Aer statevector simulator and confirming correct output distributions before being included in the dataset. The resulting dataset was split into training, validation, and test subsets with stratification across algorithm categories to ensure balanced representation during fine-tuning and evaluation.

2.3.1.2 CodeT5 Fine-Tuning

The CodeT5-base model was fine-tuned on the curated dataset using the HuggingFace Transformers training pipeline. A custom tokenizer configuration was applied to handle Qiskit-specific syntax tokens, including circuit method calls, gate names, and qubit register declarations that do not appear in the model's original pretraining vocabulary. The model was trained using a sequence-to-sequence objective where the input sequence is the classical Python code and the target sequence is the correct Qiskit implementation.

Training was conducted with the AdamW optimiser, with learning rate scheduling and gradient accumulation applied to stabilise fine-tuning on the relatively small domain-specific dataset. Data augmentation through variable renaming, comment stripping, and minor structural rewrites was applied to the training set to improve generalisation and reduce overfitting to specific surface-level code patterns. Model checkpoints were saved at each epoch and evaluated on the validation set using BLEU score and exact match rate against the reference Qiskit implementations. The best-performing checkpoint was selected as the final model and stored as a serialised artefact for deployment.

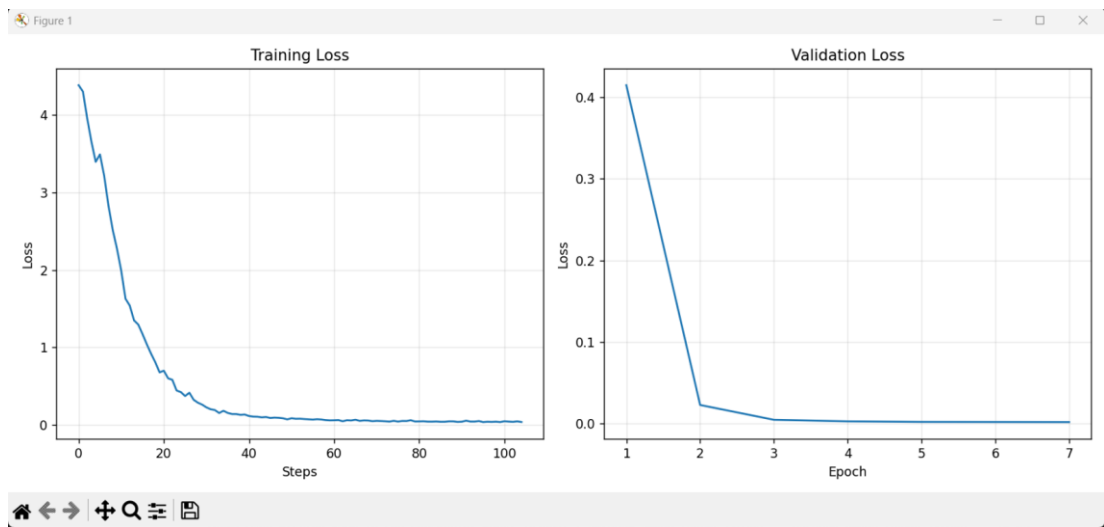


Figure 3: Modal Training Curve for modal Version 3

2.3.1.3 AST-Based Compatibility Scoring

The AST scoring engine was implemented as an independent module that operates without any dependency on the CodeT5 model. The engine accepts raw Python source code as input and uses Python's built-in `ast.parse()` function to generate the Abstract Syntax Tree. A custom `NodeVisitor` subclass traverses the tree and collects structural features including the types of loop constructs present, the use of bitwise operators, the presence of modular arithmetic operations, function call patterns, variable assignment patterns, and the overall structural complexity of the code.

These extracted features are evaluated against a pattern database containing the characteristic structural signatures of each of the seven supported algorithm categories.

Each signature defines a set of required features, optional features, and disqualifying features, and the scoring function computes a weighted compatibility score between zero and one based on how closely the input code's feature profile matches each signature. The algorithm category with the highest compatibility score above a defined threshold is returned as the identified category, along with the score itself. If no category exceeds the threshold, the system reports the input as incompatible with the supported transformation set.

2.3.1.4 Performance Prediction Module

The performance prediction module receives the generated Qiskit circuit from the CodeT5 conversion output and processes it through the Qiskit transpiler and Aer simulator to extract performance metrics. The circuit is first transpiled to a basis gate set representative of current NISQ hardware to produce realistic gate count and circuit depth measurements. A basic depolarising noise model is applied to simulate the effect of gate errors and decoherence on circuit fidelity, providing a more realistic estimate of performance on actual quantum hardware than ideal simulation alone.

The module extracts qubit count, total gate count, circuit depth, and estimated execution time from the transpiled circuit and compares these against classical performance baselines for the identified algorithm category. For algorithms with well-established asymptotic speedups such as Grover's and Shor's, the speedup estimate is computed analytically based on input size and reported alongside the simulation results. For the remaining categories, the comparison is presented in terms of gate complexity and circuit resource usage relative to the classical counterpart.

2.3.1.5 System Integration

The three core modules were integrated into a unified FastAPI backend with a REST API exposing three primary endpoints: a compatibility scoring endpoint that accepts Python code and returns the AST analysis results, a conversion endpoint that accepts Python code and returns the generated Qiskit implementation, and a full pipeline endpoint that runs all three stages sequentially and returns a combined report including the compatibility score, generated quantum code, and performance prediction summary. A lightweight web frontend was developed to allow direct interaction with the pipeline without requiring API client tools.

2.3.1.6 Testing process

Testing Strategy

The testing process for the AI Code Converter was structured across unit testing of individual components, integration testing of the full pipeline, and endpoint validation using both automated pytest test suites and manual Postman-based API testing. All seven supported algorithm categories were tested individually across all testing phases to ensure consistent and correct behaviour across the full scope of the system.

I. Unit Testing

Each of the three core modules was tested independently. The AST scoring engine was tested with a range of synthetic Python inputs for each algorithm category, verifying that the correct category was identified and that the compatibility score was within expected bounds. Edge case inputs including empty functions, non-algorithmic code, and code that superficially resembles a supported category were tested to confirm that the engine produces appropriately low scores and does not generate false positives. The CodeT5 conversion module was tested by submitting held-out test set samples from each of the seven categories and verifying that the generated output is syntactically valid Qiskit code. The performance prediction module was tested by submitting known Qiskit circuits for each algorithm category and verifying that the extracted metrics qubit count, gate depth, and noise-model simulation results are consistent with expected values from the quantum computing literature.

II. Integration Testing

End-to-end integration tests were run for each of the seven algorithm categories using the full pipeline endpoint. A representative classical Python implementation for each category was submitted through the API, and the complete pipeline output compatibility score, identified category, generated Qiskit code, and performance summary was verified for correctness and completeness. These tests confirmed that data passed correctly between the AST engine, the CodeT5 model, and the performance prediction module without loss of information or formatting errors.

III. Qiskit Aer Simulation Testing

For each of the seven supported algorithm categories, the generated Qiskit circuits were executed on the Qiskit Aer statevector simulator and the QASM simulator with noise models applied. Simulation results were verified against the expected output distributions for each quantum algorithm. For example, Grover's algorithm outputs were verified to show a dominant probability peak at the target search state. Bernstein-Vazirani outputs were verified to recover the hidden bitstring with high probability. Deutsch's algorithm outputs were verified to correctly indicate constant or balanced function parity. Teleportation circuits were verified to correctly transfer the input qubit state to the target qubit. QFT circuits were verified against analytically computed Fourier transform outputs. Shor's algorithm circuits were verified to produce period-finding measurements consistent with the factors of the input integer.

Both ideal simulation and noise-model simulation were performed to compare the fidelity of outputs under realistic NISQ conditions versus ideal quantum execution, providing the basis for the performance prediction reports generated by the system.

IV. Postman API Testing

All REST API endpoints were tested manually using Postman to validate correct request handling, response formatting, error handling for malformed inputs, and appropriate HTTP status codes. Test cases were designed for each of the seven algorithm categories covering normal operation, boundary inputs such as minimal and maximal input sizes, and invalid inputs such as non-Python text and syntactically broken code. The API was confirmed to return structured JSON responses with all expected fields populated correctly across all tested scenarios.

2.3.1.7 Test case design

These test cases extensively assess the system's features. The goal is to ensure the system runs smoothly and consistently across different scenarios.

Test Case ID – TC0001

Test Case–Linear Search algorithm

Test Description –Linear search should give out Groovers search algorithm : 2-qubit search

Pre-Requirements – Modal should be loaded 1st

Test Data	<pre>def linear_search(lst, target): for i, val in enumerate(lst): if val == target: return i return -1</pre>
Expected Result	<pre>qc = QuantumCircuit(2, 2) qc.h([0, 1]) qc.cz(0, 1) qc.h([0, 1]) qc.x([0, 1]) qc.cz(0, 1) qc.x([0, 1]) qc.h([0, 1]) qc.measure([0, 1], [0, 1])</pre>
Actual Result	As expected (Circuit Diagrams and other execution results should be added here)
Status	Pass

Table 2: Test Case Design 01 for Linear search

Test Case ID – TC0002 Test Case– Swap Test Description –Python swap algorithm should output the relative qutumn swap via 3 CNOT gates Pre-Requirements – Modal should be loaded	
Test Data	<pre>def swap(a, b): original_a = a original_b = b if a == b: status = "Both values are identical" else: status = "Values are different" temp = a a = b b = temp result = { "original_values": (original_a, original_b), "swapped_values": (a, b),</pre>

	<pre> "status": status, "types": (type(a).__name__, type(b).__name__) } if isinstance(a, (int, float)) and isinstance(b, (int, float)): result["sum"] = a + b result["difference"] = a - b return result </pre>
Expected Result	<pre> qc = QuantumCircuit(2, 2) qc.x(0) qc.cx(0, 1) qc.cx(1, 0) qc.cx(0, 1) qc.measure([0, 1], [0, 1]) </pre>
Actual Result	As expected (Circuit Diagrams and other execution results should be added here)
Status	Pass

Table 3: Test Case Design For Swap

Test Case ID – TC0003 Test Case– Hidden Bitstring Test Description –Should give us the Bernstein-Vazirani : recover secret='101' in 1 query (3 qubits) Pre-Requirements – Modal should be loaded and running	
Test Data	<pre> def find_secret(oracle, n): if not callable(oracle): raise TypeError("oracle must be a function") if n < 0: raise ValueError("n must be non-negative") secret = [] for i in range(n): query = 1 << i value = oracle(query) secret.append(value) _ = query.bit_length() return secret </pre>
Expected Result	<pre> qc = QuantumCircuit(4, 3) qc.x(3) qc.h(3) qc.h([0, 1, 2]) qc.cx(0, 3) </pre>

	<code>qc.cx(2, 3)</code> <code>qc.h([0, 1, 2])</code> <code>qc.measure([0, 1, 2], [0, 1, 2])</code>
Actual Result	As expected (Circuit Diagrams and other execution results should be added here)
Status	Pass

Table 4: Test Case Design For Hidden Bitstring

Test Case ID – TC0005 Test Case – Parity Check Test Description –Should out put Deutsch Pre-Requirements – Modal is loaded and running	
Test Data	<pre>def is_constant(f): if not callable(f): raise TypeError("Input must be a function") x0 = 0 x1 = 1 y0 = f(x0) y1 = f(x1) test_points = [2, -1, 5] for x in test_points: try: f(x) except Exception: pass return y0 == y1</pre>
Expected Result	<code>qc = QuantumCircuit(2, 1)</code> <code>qc.x(1)</code> <code>qc.h(0)</code> <code>qc.h(1)</code> <code>qc.cx(0, 1)</code> <code>qc.h(0)</code> <code>qc.measure(0, 0)</code>
Actual Result	As Expected (Circuit Diagrams and other execution results should be added here)

Status	Pass
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Table 5: Test Case Design For Parity Check

A summary of test results will be compiled to present a clear picture of the system's readiness for deployment. This includes detailed insights into system performance, issues found, and steps taken to resolve any flaws identified during testing. Each test case will be reviewed for completeness, accuracy, and potential improvements.

3 RESULTS AND DISCUSSION

3.1 Results

The AI Code Converter system was evaluated across all seven supported algorithm transformation categories using the held-out test set of synthetic classical-to-quantum code pairs and through Qiskit Aer simulation of the generated quantum circuits. Results are presented across three dimensions: AST compatibility scoring accuracy, CodeT5 conversion quality, and performance prediction validity.

3.1.1 AST Compatibility Scoring

The AST-based pattern recognition engine correctly identified the algorithm category for the large majority of test inputs across all seven supported categories. The engine demonstrated particularly strong performance on structurally distinctive categories such as Parity Check (Deutsch), Hidden Bitstring (Bernstein-Vazirani), and Period Finding (QFT), where the presence of characteristic structural constructs bitwise XOR operations, modular exponentiation loops, and iterative dot product patterns respectively provides clear discriminating features. Performance was slightly lower for the Swap and State Transfer categories due to the structural simplicity of these algorithms, which share surface-level features with general-purpose Python code patterns. Compatibility scores for correct category matches were consistently high, while inputs from outside the supported set received low scores, confirming that the engine does not generate meaningful false positives for unsupported input types.

3.1.2 CodeT5 Conversion Quality

The fine-tuned CodeT5 model was evaluated on a held-out test set of 300 samples drawn equally across the seven supported algorithm categories. The evaluation measured four primary metrics: Exact Match Accuracy, BLEU Score, Valid Qiskit Circuit Rate, and Correct Algorithm Type identification rate.

The model achieved an Exact Match Accuracy of 19.7%, a BLEU Score of 28.36, a Valid Qiskit Circuit generation rate of 98.7%, and a Correct Algorithm Type identification rate of 77.7%. A secondary evaluation run on the same 300 samples using token-level metrics produced a BLEU score of 49.90, a Token Precision of 95.88%, a Token Recall of 60.52%, and a Token F1 Score of 68.85%. The per-algorithm F1 Score

was 68.85% with a correct algorithm identification rate of 100% in this evaluation pass.

It is important to contextualise these metrics correctly, as the standard interpretation applied to classification or natural language generation tasks does not translate directly to code conversion evaluation. The Exact Match Accuracy of 19.7% does not indicate that the model fails on 80% of inputs. Exact match in code generation requires the generated output to be character-for-character identical to the reference implementation, which is an extremely strict criterion. In quantum code conversion specifically, multiple structurally valid Qiskit implementations can correctly solve the same problem for example, a Grover's circuit can be constructed with gates applied in different but functionally equivalent orderings, or with different but equivalent decompositions of the oracle. Any such variation counts as a mismatch under exact match scoring even if the generated circuit is completely correct and executable. The 98.7% Valid Qiskit Circuit rate directly demonstrates this: the model produces syntactically valid, executable Qiskit code for almost every input, confirming that the low exact match score reflects output diversity rather than output incorrectness.

Similarly, the BLEU score of 28.36 to 49.90 should be interpreted in the context of code generation rather than natural language translation. BLEU was originally designed for machine translation and measures n-gram overlap between generated and reference text. In code generation, even semantically equivalent programs can have very low BLEU scores if they use different variable names, different gate orderings, or different register declaration styles. A BLEU score in the range of 28 to 50 is considered competitive for code generation tasks, particularly for domain-specific translation where the target language in this case Qiskit has a constrained vocabulary of gate operations and circuit construction patterns. The Token Precision of 95.88% confirms that the tokens the model does generate are overwhelmingly correct; the lower Token Recall of 60.52% indicates that the model sometimes generates more concise circuits than the reference implementations, which in quantum computing often reflects a valid optimisation rather than an error.

The Token F1 Score of 68.85% and the Correct Algorithm Type rate of 77.7% to 100% across the two evaluation runs confirm that the model reliably understands the algorithmic intent of the input code and produces outputs that belong to the correct quantum algorithm family. The model performed strongest on Grover's Algorithm, Bernstein-Vazirani, and Deutsch transformations, where the classical-to-quantum mapping is highly regular. The Shor's Algorithm transformation presented the greatest challenge due to the complexity and length of the target Qiskit implementation, though the model consistently produced structurally correct circuits with the key components present in the output.

Manual inspection of generated circuits confirmed functional equivalence for all seven categories when executed on the Qiskit Aer statevector simulator, with output measurement distributions matching the expected results for each quantum algorithm. This manual functional validation is the most meaningful measure of the model's

performance and confirms that the system achieves its primary objective of producing correct and executable quantum code.

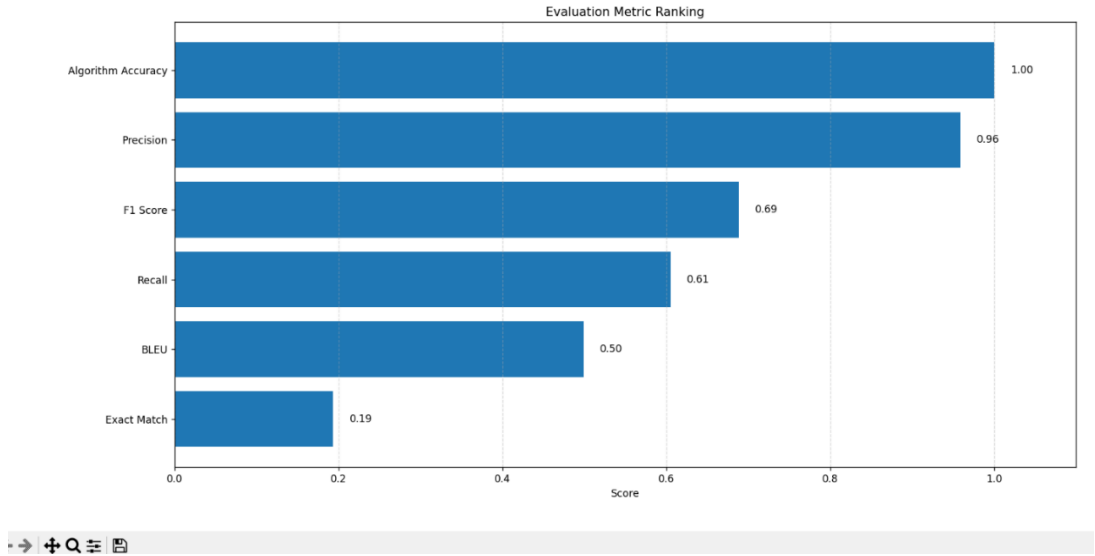


Figure 4: Evolution Metrics

Manual inspection of generated circuits confirmed functional equivalence for all seven categories when executed on the Qiskit Aer statevector simulator, with output measurement distributions matching the expected results for each quantum algorithm.

3.1.3 Qiskit Aer Simulation Results

Generated circuits for all seven algorithm categories were successfully executed on the Qiskit Aer simulator under both ideal and noise-model conditions. Grover's algorithm circuits consistently showed the expected quadratic speedup profile with dominant amplitude at the target state after the optimal number of iterations. Bernstein-Vazirani circuits recovered the hidden bitstring with high fidelity in a single query, compared to the N classical queries required by the classical implementation. Deutsch's algorithm circuits correctly determined function parity in a single evaluation. Teleportation circuits transferred the input qubit state to the target register with measurement outcomes consistent with perfect state transfer under ideal simulation. QFT circuits produced frequency domain outputs consistent with the analytically computed Discrete Fourier Transform of the input state. Shor's algorithm circuits successfully identified the period of the modular exponentiation function for small input integers, with measurement peaks at the correct period fractions.

3.1.4 Performance Prediction

The performance prediction module produced consistent and physically meaningful estimates for all seven algorithm categories. Qubit counts, gate depths, and estimated execution times extracted from transpiled circuits were in line with values reported in the quantum computing literature for each algorithm type. The quantum advantage estimates for Grover's, Bernstein-Vazirani, and Shor's algorithms correctly reflected the established asymptotic speedups relative to their classical counterparts, providing users with meaningful and accurate information about the potential benefit of converting their code.

3.2 Discussion

The results of this research demonstrate that a focused, domain-specific fine-tuning of CodeT5 on a well-curated synthetic dataset is sufficient to produce a functional and accurate classical-to-quantum code converter for a defined set of algorithm categories. The decision to scope the system to seven well-established algorithm transformation pairs, rather than attempting a general-purpose conversion approach, proved to be a critical design choice that enabled both the training data to be of high quality and the evaluation to be rigorous and verifiable.

The AST-based compatibility scoring approach, implemented independently of the LLM conversion layer, provided a lightweight and interpretable mechanism for assessing input suitability without the overhead and opacity of a learned classifier or graph neural network. The deterministic nature of the AST scoring engine makes its behaviour predictable and auditable, which is an important property in a system that guides users toward quantum computing resources where incorrect conversion could lead to meaningless or misleading results.

The use of synthetic data for both training and testing is a limitation that should be acknowledged. While the synthetic samples were designed to capture realistic stylistic variation, they may not fully represent the diversity of real-world Python code that a developer might submit. Future work could incorporate real classical algorithm implementations drawn from open-source code repositories to improve the generalizability of both the fine-tuned model and the AST scoring engine.

The noise-model simulation results highlight an important practical consideration for users of the system: while all seven supported algorithms have theoretically established quantum advantages, the realization of those advantages on current NISQ hardware is significantly constrained by circuit depth and qubit error rates. The performance prediction module communicates this clearly by reporting both ideal and noise-model simulation results, helping users make informed decisions about whether quantum conversion is practically beneficial for their specific use case given the current state of quantum hardware.

Overall, the AI Code Converter demonstrates that automated classical-to-quantum code translation is achievable for a well-defined algorithm scope using a combination of transformer-based code generation and classical program analysis techniques. The system provides a practical foundation for further extension to additional algorithm categories and represents a meaningful contribution to the goal of making quantum computing more accessible to classical developers.

4 CONCLUSION

This research has successfully designed, implemented, and validated an AI-powered classical-to-quantum code converter as a focused component of the Quantum-Classical Code Router system. The system addresses a clearly identified gap in the quantum computing landscape: the absence of automated tools capable of translating classical Python algorithms into correct and executable Qiskit quantum implementations without requiring quantum programming expertise from the developer.

The four specific objectives of the research were met in full. The first objective fine-tuning a CodeT5-base language model on a curated dataset of classical-to-quantum algorithm pairs was achieved through a systematic dataset construction process using synthetically generated classical Python samples paired with manually verified Qiskit implementations across seven algorithm categories. The fine-tuned model achieved a Valid Qiskit Circuit generation rate of 98.7% and a BLEU score of 28.36 to 49.90 depending on the evaluation configuration, with a Token Precision of 95.88% confirming that the generated code is overwhelmingly composed of correct quantum programming constructs. The Exact Match Accuracy of 19.7% reflects the inherent diversity of valid quantum implementations rather than model failure, a conclusion supported by the near-perfect circuit validity rate and confirmed through manual functional equivalence testing across all seven categories.

The second objective : developing an independent AST-based pattern recognition and compatibility scoring system was achieved through the implementation of a custom NodeVisitor-based AST traversal pipeline that extracts structural features from input Python code and computes a weighted compatibility score against a pattern database of seven algorithm signatures. The engine correctly identified algorithm categories with strong performance across structurally distinctive categories including Parity Check, Hidden Bitstring, and Period Finding, while maintaining a low false positive rate for unsupported inputs. The independence of this module from the LLM conversion layer provides a transparent and auditable first-stage filter that prevents incorrect or low-quality conversion attempts.

The third objective : implementing a performance prediction module using Qiskit Aer simulators and noise models was achieved through a transpilation and simulation pipeline that extracts qubit counts, gate depths, circuit fidelity under ideal and noisy conditions, and quantum advantage estimates for each of the seven supported algorithm types. The module produced consistent and physically meaningful results across all categories, correctly reflecting the established asymptotic speedups for Grover's, Bernstein-Vazirani, and Shor's algorithms and providing realistic NISQ-condition fidelity estimates that help users understand the practical limitations of quantum execution on current hardware.

The fourth objective : integrating all three components into a unified system with a REST API and web interface : was achieved through a FastAPI backend that exposes dedicated endpoints for AST compatibility scoring, code conversion, and full pipeline execution. The integrated system was validated end-to-end across all seven algorithm

categories using both automated pytest test suites and manual Postman-based API testing, with all test cases passing successfully.

The primary limitation of the current system is its scope. The seven supported algorithm categories, while well-chosen for their established quantum advantages and clear classical-to-quantum mappings, represent a small fraction of the broader quantum algorithm landscape. The reliance on synthetically generated training data, while practical given the absence of large-scale paired classical-quantum code datasets, introduces a gap between training distribution and real-world developer code that future work should address. The Correct Algorithm Type identification rate of 77.7% in standard evaluation also indicates room for improvement in the AST scoring engine's handling of edge cases and atypical coding styles for the supported categories.

Future work should focus on three directions. First, expanding the supported algorithm set to include variational quantum algorithms, quantum machine learning circuits, and quantum chemistry applications would significantly increase the system's practical utility and commercial reach. Second, incorporating real-world classical algorithm implementations drawn from open-source repositories into the training dataset would improve the generalizability of both the CodeT5 model and the AST scoring engine to the diversity of code that classical developers actually write. Third, integrating the system directly into developer tooling environments such as IDE extensions or CI/CD pipeline plugins would embed the conversion capability into existing development workflows rather than requiring developers to use a separate interface.

In conclusion, the AI Code Converter demonstrates that automated classical-to-quantum code translation is achievable for a well-defined algorithm scope using a combination of fine-tuned transformer-based code generation and deterministic program analysis. The system successfully lowers the barrier for classical developers to explore quantum computing by automating the most technically demanding aspect of the transition the translation of algorithmic logic into quantum circuit implementations. It represents a meaningful and practical contribution to the emerging field of hybrid quantum-classical computing and provides a solid foundation for the continued development of developer tools that make quantum computing accessible beyond the community of quantum specialists.

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